

Core Relationships: $\vec{E}$, V , and I

Electric Field $\vec{E}$: force per unit charge; points from + to –.
Potential V : energy per unit charge (scalar).
Current I : charge flow per unit time.

Moving between them:

$$V_B - V_A = - \int_A^B \vec{E} \cdot d\vec{l} \quad (\text{path independent})$$

$$\vec{E} = -\nabla V \quad (1\text{-D shortcut: } E_x = -dV/dx)$$

$$I = \frac{dq}{dt} = nqAv_d \quad (\text{Microscopic Current})$$

n : carrier density, A : area, v_d : drift speed

$$v_d = \frac{I}{nqA} \quad (\text{drift speed; typically } \sim \text{mm/s})$$

$$\vec{J} = nq\vec{v}_d \quad (\text{current density [A/m}^2\text{)])}$$

Key signs: $\Delta V > 0$ moving against $\vec{E}$; $\Delta V < 0$ with $\vec{E}$.
 Work by $\vec{E}$: $W = q\Delta V = q(V_i - V_f)$.

Ch. 16 – Electric Potential & Potential Energy

Potential is a scalar; superpose by addition, not vector sum.

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (\text{point charge})$$

$$U = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} \quad (\text{system PE})$$

$$\Delta V = -E \Delta x \quad (\text{uniform field})$$

$$V = - \int \vec{E} \cdot d\vec{l} \quad (\text{add contributions})$$

“Electric Potential” $\neq$ “Electric Potential Energy”:
 $U = qV$ (energy); V alone is the potential (energy/charge).

Ch. 16 – V from $\vec{E}$ in Non-Uniform Fields

Break path into pieces; $\vec{E}$ may vary along the path.

$$\Delta V = - \int_i^f \vec{E} \cdot d\vec{l} \quad (\text{path independent})$$

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \quad (\text{scalar, easier than } \vec{E})$$

Steps: **(1)** choose convenient path, **(2)** express $\vec{E}(l)$,
(3) evaluate $\int \vec{E} \cdot d\vec{l}$, **(4)** $\Delta V = -\text{result}$.

Ch. 17 – Electric Field in Circuits

Mobile charges redistribute until steady state; the surface charge gradient creates $\vec{E}$ inside.

$$\vec{J} = \sigma \vec{E} \quad (\text{Ohm, microscopic})$$

$$V = IR, \quad R = \frac{\rho L}{A}, \quad \rho = \frac{1}{\sigma} \quad [\Omega \cdot \text{m}]$$

$$v_d = \frac{I}{nqA}, \quad |\vec{E}| = \frac{V}{L} \quad (\text{uniform})$$

In steady state: $\vec{E}_{\perp \text{surface}} = 0$ inside conductor; charge resides on surface.

Ch. 18 – Circuits: Loop Rule & Kirchhoff

Closed loops conserve energy; junctions conserve charge.

$$\sum_{\text{loop}} \Delta V = 0 \quad (\text{KVL – Loop rule})$$

$$\sum I_{\text{in}} = \sum I_{\text{out}} \quad (\text{KCL – Junction rule})$$

$$R_{\text{eq}} = \sum R_i \quad (\text{series})$$

$$\frac{1}{R_{\text{eq}}} = \sum \frac{1}{R_i} \quad (\text{parallel})$$

$$P = IV = I^2 R = \frac{V^2}{R} \quad [\text{W}]$$

Sign convention: $-IR$ across resistor (in current direction); $+\mathcal{E}$ across EMF (negative to positive terminal).

Ch. 19 – RC Circuits & Capacitors

Capacitors store charge; RC circuits decay/grow exponentially.

$$C = \frac{Q}{V}, \quad C = \frac{\epsilon_0 A}{d} \quad (\text{parallel-plate})$$

$$U = \frac{1}{2} CV^2 = \frac{Q^2}{2C} \quad (\text{energy stored})$$

$$Q(t) = C\mathcal{E}(1 - e^{-t/\tau}) \quad (\text{charging})$$

$$Q(t) = Q_0 e^{-t/\tau} \quad (\text{discharging})$$

$$\tau = RC, \quad I(t) = \frac{\mathcal{E}}{R} e^{-t/\tau}$$

Series: $\frac{1}{C_{\text{eq}}} = \sum \frac{1}{C_i}$ Parallel: $C_{\text{eq}} = \sum C_i$

Constants & Quick Reference

$$\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$$

$$e = 1.6 \times 10^{-19} \text{ C}, \quad m_e = 9.1 \times 10^{-31} \text{ kg}$$

Quantity	SI Unit
V (potential)	$V = \text{J/C}$
E (field)	$\text{N/C} = \text{V/m}$
I (current)	$A = \text{C/s}$
R (resistance)	$\Omega = \text{V/A}$
C (capacitance)	$F = \text{C/V}$
ρ (resistivity)	$\Omega \cdot \text{m}$
P (power)	$W = \text{J/s}$

Insulator vs. Conductor in $\vec{E}$:

Conductor (static): $\vec{E}_{\text{inside}} = 0$, $V = \text{const.}$

Insulator: $\vec{E} \neq 0$ inside; $\Delta V = \int \vec{E} \cdot d\vec{l} \neq 0$.

$$E_x = -\frac{dV}{dx} \quad (\text{gradient shortcut, 1-D})$$